Muhammad Usman

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Skills

- Programming Languages: C++ | Python | Javascript | C |
- Framworks / Tech Stack: React |Node-JS | Express | NextJS | MongoDB | HTML/CSS |
- Basic Artificial Intelligence
- Others: Git | GitHub | Agile | Jira |

Education

Bachelor of Computer Science	**FAST-NUCES**	*Faisalabad, Pakistan*	**22/2022 - Present**
Intermediate in Computer Science	**Punjab College of Commerce**	*Faisalabad, Pakistan*	**2020 - 2022**

Projects

- **Nodex-p2p-communication-app** Developed a secure P2P chat application with a GUI using Python and Tkinter, supporting user authentication, encrypted messaging (Fernet), private/group chats, file transfer, emoji integration, and real-time message deletion. Enabled interactive communication with features like message IDs and file download notifications across clients.(2024)link
- **FlexiHome – Furniture Rental Web App** Developed a responsive furniture rental website where vendors can list furniture for rent and customers can browse and book items. Completed the full frontend, including user interface, vendor listing layout, and booking flow. Backend integration planned for future development. (2025) link
- **Mini OS Simulation (C++ & Shell Scripting)** – Developed a lightweight command-line based operating system simulation featuring mini applications such as Calculator, Notepad, Calendar, and simple games. Implemented core functionalities using C++ and Bash scripting to simulate OS-level operations like file management, system commands, and multitool navigation.link
- **Transport Management System – C++ (Desktop Application)**
  Developed a file-based transport management application using C++ and Visual Studio, supporting vehicle tracking, service booking, and user authentication. Implemented modular components for managing drivers, customers, and maintenance records through structured header and source files. Designed the system to handle real-time operations with persistent data storage in text files.link
- **The Algorithms (Python)(Open Source)** Contributed a Python implementation of the Count Inversions algorithm using a modified merge sort.Wrote clean, tested, and documented code following community standards and style guidelines.Collaborated through GitHub PRs and passed automated CI checks. (2024) link
- **AI-Based Game Simulations (Python)**Developed classic games like Checkers, Tic Tac Toe, and N-Queens using AI algorithms including Hill Climbing, Simulated Annealing, Local Beam Search, and Minimax with Alpha-Beta Pruning. Implemented game logic and UI using Pygame, showcasing practical applications of search algorithms and heuristic strategies.link
- **Sentiment Analysis System (Python):**
  Developed a sentiment analysis application to classify text data as **positive, negative, or neutral** using **Natural Language Processing (NLP)** techniques. Implemented text preprocessing, feature extraction, and machine learning models using **Python libraries such as NLTK, Scikit-learn, and Pandas**, demonstrating practical application of **text analytics and data-driven decision making**. (link)
- **Pseudo Code to C++ Generator (Python, NLP)**
  Fine-tuned a **GPT-2 language model using LoRA** to automatically convert **pseudo-code into C++ source code**. Developed an interactive **Streamlit web application** for user input and real-time code generation. Integrated the **Hugging Face Transformers** library for model loading and inference, implemented configurable generation parameters, and enabled **code download functionality**, demonstrating practical experience in **LLM fine-tuning, prompt engineering, and deployment**. (link)
- **Urdu Text Transformer (Python, PyTorch, NLP)**
  Developed a **custom Transformer model from scratch** for **Urdu text processing**, implementing **encoder-decoder architecture, multi-head attention, positional encoding, and feed-forward layers**. Built **tokenization, vocabulary management, and text normalization pipelines** to handle Urdu-specific characters and diacritics. Integrated **greedy decoding for sequence generation** and designed a **modular PyTorch framework** for training and inference, demonstrating deep understanding of **transformer architectures, sequence modeling, and multilingual NLP**. (link)
- **Text Summarization with T5 (Python, PyTorch, Transformers, Streamlit)**
  Developed an **interactive web application** for generating concise summaries from long-form text using a **fine-tuned T5-small model** trained on the CNN/DailyMail dataset. Implemented **beam search decoding with length constraints and repetition penalties**, automated **input preprocessing with 'summarize:' prefix**, and a **Streamlit-based UI** featuring dynamic summary length sliders, original text preview, and compression ratio metrics. The project demonstrates practical application of **transformer-based NLP models for real-world summarization tasks**. (link)

- **Seq2Seq Character-Level Translator (Python, PyTorch)**
  Implemented a **bidirectional LSTM encoder-decoder model** with **multi-layer decoder** and **layer projection** to handle mismatched encoder-decoder depths. Designed for character-level sequence-to-sequence tasks with **teacher forcing**, dropout regularization, and dynamic handling of variable-length input sequences. Developed modular **Encoder, Decoder, and Seq2Seq classes** with robust forward pass logic, including **SOS/EOS token management** and **dynamic teacher forcing**. Verified model with **dummy tensor tests** to ensure correct tensor shapes and end-to-end functionality.(link)
- **Clinical Retrieval-Augmented Generation (RAG) System**
  Built a **medical question-answering system** using **FAISS for vector retrieval**, **SentenceTransformers embeddings**, and **Llama-3.1-8B via Groq API**. Designed a **strict prompt-guided LLM pipeline** that only uses explicit clinical notes, enforcing fact-based responses and avoiding assumptions. Implemented a **Streamlit UI** for interactive querying, displaying retrieved documents with evidence-based summaries, and generating precise answers. Optimized retrieval and generation with caching and embedding normalization for real-time performance.(link)

Certifications

- **JavaScript (Intermediate) Certificate:** Javascript Intermediate Certificate. **(02/2023 - Persent)**
- **JavaScript (Fundamentals) Certificate:** Javascript Fundamentals Certificate. **(02/2023 - Present)**
- **Git/Github(Intermediate) Certificate:** Git/Github intermediate Certificate.**(04/2025-Present)**
- **Microsoft Office (intermediate) Certificate:** Microsoft office intermediate Certificate.**(04/2021-09/2021)**

Experience

Front-End WEB INTERN **Tech-Feast-JBS** *Faisalabad, Pakistan* **NOV 2021 - FEB 2022**

- Worked on front-end development using **Bootstrap** and **HTML/CSS**
- Worked on search engine optimization
- Collaborated with cross-functional teams to deliver high-quality products within project timelines.
- Worked in an Agile software development environment to deliver high-quality products.

Front-End Web Intern **WriteO AI** *Lahore,Pakistan* **June 2025 - Oct 2025**

- Built interactive front-end features using **React.js**
- Developed an **endless canvas** feature for dynamic content creation
- Implemented **AI-powered image generation** using the **Gemini API**
- Created a **multi-purpose to-do list application** with modern UI/UX principles
- Collaborated with the team to integrate APIs and improve user experience

Others

- **Daira Marketing Team:** Collaborated with the Daira Marketing Team to execute marketing strategies for the event. **(2025)**
- **FAST Gaming Society**: I contributed to the Gaming Society by hosting tekken-7 event. **(2024 - present)**